Guillermo Serrera

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Employment

2023–26	**Predoctoral researcher**, University of Cantabria

Education

University education

2022–26*	**Ph.D.**, Science and Technology, University of Cantabria
2021–22	**M.Sc.**, Science and Engineering of Light, University of Cantabria
2016–21	**B.Sc.** Physics, University of Cantabria

Other training

2026	Motivation and learning (8 hours), University of Cantabria courses.
2026	Reflection on the own teaching practice in higher education (9 hours), University of Cantabria courses.
2026	The revolution of generative artificial intelligence in research and teaching (12 hours), University of Cantabria courses.
2025	Strategies for Including the SDGs in university teaching (3 hours), University of Cantabria courses.
2025	ACS Reviewer Lab (2 hours), American Chemical Society.
2025	Optica Publishing Group Certified Reviewer Course (1 hour), Optica Society of America.
2025	Fundamentals and applications of optical materials (20 hours), University of Cantabria summer school.
2024	Certified Peer Reviewer Course (4 hours), Elsevier Researcher Academy.
2023	Creative strategies in teaching and research (6 hours), University of Cantabria courses.
2023	Social competences: tools for university teachers (6 hours), University of Cantabria courses.
2023	Emotional competences: tools for university teachers (6 hours), University of Cantabria courses.
2022	European School on Plasmonics and Phase Change Materials (16 hours), PHEMTRONICS H2020 FET Open project.

Research Experience

2023–26	Researcher in the project “Modelling of the Photothermal response of hybrid nanoSystems (MOPHOSYS)” Advisor: Pablo Albella (University of Cantabria), Spain.

*Expected.

2021–22	Part time researcher in the project "Spectroscopy of planetary and atmospheric particulae by optical tweezers". Advisor: Onofrio Maragò (IPCF-CNR), Italy.

Awards, Funding & Honors

2025	Funding for short research stays by the Spanish Ministry of Science, Innovation and Universities (EST25/00316).
2025	Best theoretical poster at CEN2025 conference.
2023	Best poster at University of Cantabria Science day.
2023–26	University Professorship Training (FPU21/02296) Fellow at the University of Cantabria. Funded by the Spanish Ministry of Science, Innovation and Universities.
2020–21	Collaboration Research Grant at the Applied Physics Department of the University of Cantabria. Funded by the Spanish Ministry of Science, Innovation and Universities.

Publications

G Google Scholar

Journal Articles

J1. **Guillermo Serrera**, Gutiérrez, Y., Gil, J. J. & Moreno, F. "Stokes-Based 3D Polarimetric Analysis of Dipolar Near-Fields". *Advanced Photonics Research* **7,** e70251. https://doi.org/10.1002/adpr.70251 (2026).

J2. **Guillermo Serrera** & Albella, P. "Enhanced Enantioselective Optical Trapping enabled by Longitudinal Mie Resonances in Silicon Nanodisks". *Nanophotonics* **15,** e70207. https://doi.org/10.1002/nap2.70207 (2026).

J3. Sanz-Paz, M., Siegel, N., **Guillermo Serrera**, González-Colsa, J., Zhu, F., Kołątaj, K., Fujii, M., Sugimoto, H., Albella, P. & Acuna, G. P. "Color Routing and Beam Steering of Single-Molecule Emission with a Spherical Silicon Nanoantenna". *Advanced Functional Materials* **36,** e29955. https://doi.org/10.1002/adfm.202529955 (2026).

J4. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. "3-dimensional plasmonic nanomotors enabled by independent integration of optical pulling and lateral forces". *Nanophotonics* **14,** 2500299. https://doi.org/10.1515/nanoph-2025-0374 (2025).

J5. Siegel, N., Sanz-Paz, M., González-Colsa, J., **Guillermo Serrera**, Zhu, F., Szalai, A. M., Kołątaj, K., Fujii, M., Sugimoto, H., Albella, P. & Acuna, G. P. "Distance-Dependent Interaction between a Single Emitter and a Single Dielectric Nanoparticle Using DNA Origami". *Small Structures* **6,** 2500299. https://doi.org/10.1002/sstr.202500299 (2025).

J6. **Guillermo Serrera**, González-Colsa, J. & Albella, P. "Amplified linear and nonlinear chiral sensing assisted by anapole modes in hybrid metasurfaces". *Applied Physics Letters* **124,** 251701. https://doi.org/10.1063/5.0212393 (2024).

J7. **Guillermo Serrera**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. "Enhanced optical chirality with directional emission of Surface Plasmon Polaritons for chiral sensing applications". *Journal of Quantitative Spectroscopy and Radiative Transfer* **284,** 108166. https://doi.org/10.1016/j.jqsrt.2022.108166 (2022).

J8. González-Colsa, J., **Guillermo Serrera**, Saiz, J. M., Ortiz, D., González, F., Bresme, F., Moreno, F. & Albella, P. "Gold nanodoughnut as an outstanding nanoheater for photothermal applications". *Optics Express* **30,** 125–137. https://doi.org/10.1364/OE.446637 (2022).

J9. González-Colsa, J., **Guillermo Serrera**, Saiz, J. M., González, F., Moreno, F. & Albella, P. "On the performance of a tunable grating-based high sensitivity unidirectional plasmonic sensor". *Optics Express* **29,** 13733–13745. https://doi.org/10.1364/OE.422026 (2021).

Conference contributions

Talks

T1. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Light-Driven Nanomotors with controllable 3D Motion.* SPIE Optics + Photonics 2026 (San Diego, United States). Aug. 2026.

T2. **Guillermo Serrera**, Godiksen, R. H., Dedding, L. M., Sistermans, T. T. C., Mohammadi, E., Raziman, T. V., S., R., Albella, P. & Curto, A. G. *Circularly polarized Raman scattering for optical spin detection in silicon metasurfaces.* Invited talk at META 2026 (Dublin, Ireland). July 2026.

T3. **Guillermo Serrera**, Godiksen, R. H., Dedding, L. M., Sistermans, T. T. C., Mohammadi, E., Raziman, T. V., S., R., Albella, P. & Curto, A. G. *Probing optical spin with circularly polarized Raman scattering in silicon metasurfaces.* Invited talk at 14th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2026) (Marbella, Spain). Mar. 2026.

T4. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Control of 3-Dimensional Motion in Plasmonic Nanomotors with Optical Pulling Forces.* Invited talk at META 2025 (Málaga, Spain). July 2025.

T5. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Optical pulling forces for 3-dimensional motion control in plasmonic nanomotors.* 13th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2025) (Porto, Portugal). Feb. 2025.

T6. **Guillermo Serrera**, González-Colsa, J. & Albella, P. *Anapole modes enabling linear and nonlinear chiral sensing in hybrid metasurfaces.* META 2024 (Toyama, Japan). July 2024.

T7. **Guillermo Serrera**, González-Colsa, J. & Albella, P. *Hybrid metal-dielectric resonator for anapole-assisted enhanced chiral sensingapplications.* SPIE Photonics Europe (Strasbourg, France). Apr. 2024.

T8. **Guillermo Serrera**, González-Colsa, J. & Albella, P. *Exploiting Hybrid Metasurfaces for enhanced Linear and Nonlinear Chiral Sensing Applications.* 12th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2024) (Rome, Italy). Feb. 2024.

T9. **Guillermo Serrera**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. *Chiral sensing based on directional plasmons.* International Conference of Quantum, Nonlinear and Nanophotonics 2022 (ICQNN'22) (Jena, Germany). Sept. 2022.

T10. **Guillermo Serrera**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. *Directional Plasmon Emission in a Mixed Dielectric-plasmonic Structure: Application to Chiral Sensing.* 10th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2022) (Online). Feb. 2022.

Posters

P1. **Guillermo Serrera**, Gutiérrez, Y., Gil, J. J. & Moreno, F. *3D Stokes Polarimetry of Dipolar Near-Fields.* META 2026 (Dublin, Ireland). July 2026.

P2. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Enabling light-driven 3D control of nanomotors with optical forces.* Researcher's Tuesday (Mons, Belgium). Mar. 2026.

P3. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Innovative approaches to 3D motion control in plasmonic nanomotors with optical pulling forces.* CEN2025 (Madrid, Spain). June 2025.

P4. **Guillermo Serrera**, González-Colsa, J., González, F., Moreno, F., Saiz, J. M. & Albella, P. *Grating metasurfaces as directional plasmon sources: applications in achiral and chiral sensing.* ImagineNano 2021 - 3PM2021 (Bilbao, Spain). Nov. 2021.

P5. **Guillermo Serrera**, González-Colsa, J., González, F., Moreno, F., Saiz, J. M. & Albella, P. *Enhanced chiroptical activity with dielectric chiral nanostructures: applications in racemic-mixture sensing.* International Conference on Metamaterials and Nanophotonics (Metanano 2021) (Online). Sept. 2021.

Softwares

S1. **Guillermo Serrera**, Gutiérrez, Y. & Gil, J. J. *DiPolIn (DIpolar POLarization in INtrinsic reference frame)* Free software uploaded to Zenodo (DOI: 10.5281/zenodo.18682022). Feb. 2026.

Teaching

More than 170 hours of teaching in the following courses and institutions:

University of Cantabria

2025–26	Teaching Assistant, Physics I (G375)
2024–26	Teaching Assistant, Physics II (G383/G393)
2023–26	Teaching Assistant, Advanced Experimental Techniques (G1775)
2023–24	Teaching Assistant, Energy in the World of Today (G1675)

Supervisions

Undergraduate

2026	Víctor Johan Moreno Cedeño, University of Cantabria, *A polarimetric study of multipolar scattering in arbitrary nanoscatterers.*
2024	Bárbara Cacicedo López, University of Cantabria, *Study of structured light scattering by nanostructures: application in optical chirality.*

Research stays and visits

2025	University of Fribourg, two-month research stay at the Photonic Nanosystems Group, supervised by Prof. Guillermo P. Acuña.
2024	Ghent University, three-month research stay at the Photonics Research Group, supervised by Prof. Alberto G. Curto.

2021	Instituto de Estructura de la Materia, CSIC, visit to Dr. Vincenzo Giannini, attended the seminars “Novel plasmonic metamaterials: applications in sensing”, and “Thermal effects in nanoparticles”.
2020	University of Oviedo, visit to Prof. Pablo Alonso-González, attended the seminar “Advanced near-field optics”.

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